

Pre-Teen Safety Smartwatch

*Design and Development of a Pre-Teen Safety Smartwatch Using ESP32 and
Cellular Connectivity*

Destiny Nosa - 22775725

EE423 Project



A thesis submitted in partial fulfilment of the requirements for the Maynooth University
ME in Electronic Engineering

December 2025

Supervisor: Andrew Meehan

Abstract

This project presents the design and prototyping of a low-cost safety smartwatch aimed at pre-teen users. The system combines an ESP32-S3 microcontroller, an LTE-capable cellular modem, a 0.96" OLED display, user-input buttons, and a rechargeable Li-ion power system. The objective is to create a wearable device that supports location reporting, emergency messaging, and simple on-device interaction while maintaining low power consumption and compact physical dimensions.

The early development phase involved building a complete Fritzing schematic, followed by a breadboard prototype to validate the core subsystems, including the ESP32, OLED I²C communication, button inputs, and battery voltage sensing. Cellular communication was not tested on the breadboard due to the modem's pin configuration, making PCB integration the preferred approach.

A full custom PCB was designed in EasyEDA Pro, including controlled routing, ground-plane optimisation, and component placement suited to smartwatch enclosure constraints. The PCB was then prepared for manufacturing, and an updated 3D enclosure model is being developed to fit the new board shape and component locations.

This report outlines the technical background, design decisions, implementation steps, and lessons learned throughout the prototyping process. Future work includes full firmware development, safety-feature integration, enclosure finalisation, and real-world testing.

Declaration

I declare that this thesis and the project it describes are exclusively my own work, except where specifically noted. I confirm that I have appropriately cited all information derived from the published and unpublished work of others.

Table of Contents

Abstract.....	1
Declaration	2
1 Introduction	4
1.1 Original Contributions and Externally Provided Resources	5
1.2 Structure of report.....	5
2 Technical Background	6
3 Body of the work	9
3.1 Breadboard Prototyping.....	9
3.2 PCB Schematic and Layout (EasyEDA Pro)	10
3.3 Soldering and PCB Assembly	11
3.4 Enclosure Development.....	12
3.5 Firmware Progress.....	13
4 Conclusions and future work.....	15
Glossary of terms	16
References	16
Appendix.....	17
Risk Assessment.....	18

1 Introduction

This project explores the design and implementation of a custom smartwatch targeted at pre-teen users, with features focused on safety, communication, and ease of use. Unlike commercial smartwatches, this design emphasises simplicity, low cost, and customisability, making it suitable for research, prototyping, and potential educational or assistive applications.

The project integrates several engineering domains: embedded systems, PCB design, low-power electronics, wireless communication, and mechanical enclosure design. The smartwatch combines an ESP32-S3 microcontroller, a SIM-enabled modem for SMS and data connectivity, a small OLED display for UI, push buttons for user input, and a battery-charging and power-management system.

1.1 Original Contributions and Externally Provided Resources

Original contributions

- Complete schematic and block-level hardware architecture created using Fritzing and EasyEDA.
- Breadboard prototype demonstrating ESP32 functionality, OLED communication, button testing, and battery monitoring.
- Custom PCB design including manual routing, ground-plane optimisation, decoupling placement, power distribution, and modem interface.
- Initial 3D enclosure design, redesigned after PCB creation to ensure mechanical compatibility.
- Development of an early firmware template integrating display control, input handling, and GSM initialisation.

Externally provided resources

- SIMCom / Quectel modem datasheets and application notes.
- Libraries for OLED (SSD1306), ESP32 board package, and Arduino GSM libraries.
- Fritzing/EasyEDA component libraries for schematic symbols and PCB footprints.
- Supervisor and other lecturers' guidance and feedback on design approach.

1.2 Structure of report

The remainder of this report is organised as follows:

Section 2 — Technical Background

Overview of ESP32 architecture, GSM/LTE modems, battery management, PCB design practices, and smartwatch design constraints.

Section 3 — Body of the Work

Breadboard prototyping, schematic creation, PCB design, routing decisions, enclosure development, and partial firmware progress.

Section 4 — Conclusions and Future Work

Summary of achievements and the next steps toward full smartwatch functionality.

2 Technical Background

This section presents the key technical concepts and design principles underpinning the development of the pre-teen smartwatch prototype. It outlines the operation of the ESP32-S3 microcontroller, cellular communication using an LTE-M/NB-IoT modem, wearable power management, OLED display interfacing, PCB layout considerations, and mechanical constraints of compact enclosures. Understanding these topics is essential to justify the hardware and software design decisions made, ensure reliable system operation, and demonstrate that the final implementation meets performance, safety, and usability requirements for a wearable IoT device.

ESP32 Architecture and Low-Power Wireless Capabilities

The ESP32-S3 microcontroller is a dual-core system-on-chip integrating Wi-Fi and Bluetooth Low Energy, making it suitable for compact and battery-powered wearable devices. Its low-power modes, GPIO flexibility, and native support for peripherals such as I²C and UART enable efficient integration with displays, sensors, and cellular modems while maintaining low energy consumption in idle states. [1]

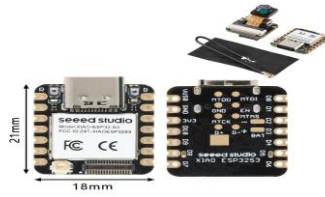


Figure 1 Seed Studio XIAO ESP32-S3 development board showing compact form factor and pin layout used as the main microcontroller in the smartwatch prototype.

LTE-M / NB-IoT / GSM Cellular Modem Fundamentals

The NL-SW-LTE-QBG95 cellular modem provides LTE-M, NB-IoT, and 2G GSM connectivity, allowing reliable SMS and data transmission with reduced power consumption compared to traditional LTE. Communication with the microcontroller is achieved via UART using AT commands, while integrated GNSS enables location tracking for safety-focused wearable applications. [2]



Figure 2 SIMCom cellular modem module used for LTE-M / NB-IoT / GSM communication, enabling SMS transmission and network connectivity.

Power Supply Design for Wearables (Li-ion, TP4056, VBAT Sensing)

Wearable devices commonly use single-cell 3.7 V Li-ion or Li-Po batteries due to their high energy density and compact size. Charging and protection are handled by a TP4056 module, while VBAT sensing through a resistor divider allows the microcontroller to monitor battery voltage and prevent deep discharge. Proper decoupling capacitors are required to handle current spikes from the cellular modem.

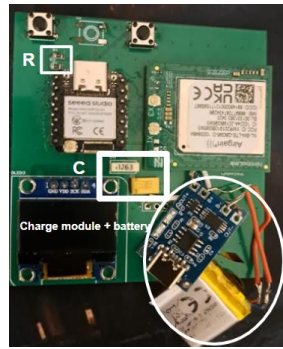


Figure 3 Power subsystem comprising the Li-ion battery and charging module, responsible for portable power supply and battery management.

OLED I²C Communication Principles

The OLED display communicates with the ESP32 via the I²C protocol using shared SDA and SCL lines. This interface allows low pin usage and reliable short-range communication, making it ideal for small PCBs. The display provides immediate visual feedback to the user, confirming message selection and successful transmission.



Figure 4: 0.96-inch OLED display module used to provide visual feedback and status information to the user.

PCB Grounding, Decoupling, and Layout Considerations

Proper grounding and decoupling are critical in mixed-signal designs combining RF, digital, and power electronics. A common ground plane reduces noise, while local decoupling capacitors near power pins stabilize voltage during transient current demands, particularly from the cellular modem during transmission.

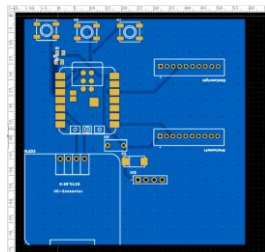


Figure 5 Custom PCB layout design showing component placement, signal routing, and connector interfaces for the smartwatch system.

Mechanical Constraints of Smartwatch Enclosures

Smartwatch enclosures impose strict size and height constraints, directly influencing component selection and PCB layout. Components must be low-profile, securely mounted, and mechanically robust to ensure reliable operation within a compact, wearable form factor.

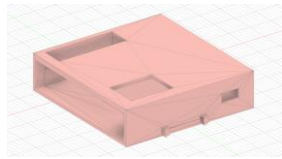


Figure 6 3D CAD model of the smartwatch enclosure, designed to house the PCB, battery, and display within compact mechanical constraints.

3 Body of the work

3.1 Breadboard Prototyping

The first stage of development focused on component selection and validating the core circuit functionality using a breadboard prototype. A complete circuit layout was first constructed in Fritzing to clearly define all electrical connections and ensure consistency with the intended schematic.

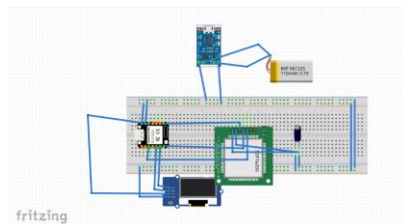


Figure 7. Fritzing schematic of the breadboard prototype, illustrating power distribution, I²C connections to the OLED display, button inputs, and preliminary battery voltage sensing circuitry.

The ESP32-S3 microcontroller was then tested using the Arduino IDE to confirm successful programming, power delivery, and serial communication. Basic test sketches were uploaded to verify correct boot behaviour and GPIO functionality before integrating additional peripherals.

The OLED display was validated separately by scanning the I²C bus to confirm device detection, followed by running example display code to ensure correct communication and screen

operation. Button inputs were also tested, with software debouncing implemented to prevent false triggering during presses.

Finally, the battery voltage divider circuit was tested by measuring the scaled battery voltage using the ESP32’s ADC, confirming accurate voltage sensing. The SimCom cellular modem was not tested at the breadboard stage due to not having an available Sim card.

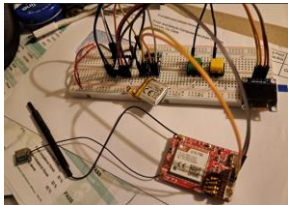


Figure 8. Breadboard prototype of the pre-teen smartwatch system, showing the ESP32-S3 microcontroller, OLED display, tactile buttons, Li-ion battery, and cellular modem module connected for early functional testing.

3.2 PCB Schematic and Layout (EasyEDA Pro)

Following successful breadboard validation, the circuit was transferred into **EasyEDA Pro** with a few changes in the SimCom (I bought a different one), resistors and capacitors, then I develop the PCB schematic and layout. All functional blocks including the ESP32-S3 microcontroller, OLED display, push buttons, power circuitry, and SIMCom cellular modem were accurately recreated in the schematic environment to reflect the tested prototype.

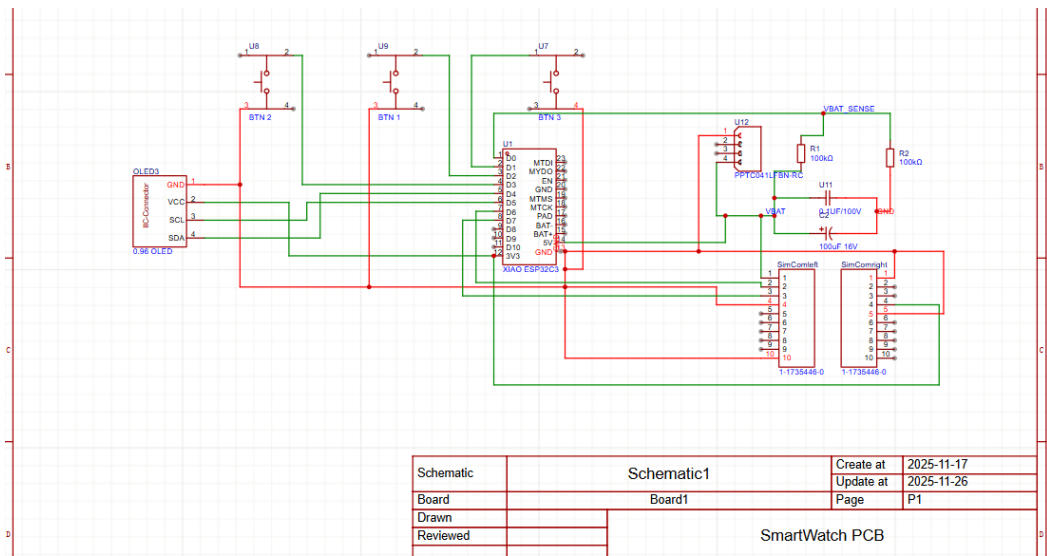


Figure 9. EasyEDA schematic of the smartwatch PCB showing the ESP32-S3, OLED display, buttons, battery sensing circuit, and SIMCom modem connections.

To improve readability and reduce wiring complexity, **net labels** were extensively used for key signals such as *SCL*, *SDA*, *BTN1*, *BTN2*, *VBAT*, and *VBAT_SENSE*. This approach ensured consistent signal naming across the design and reduced the risk of routing errors during layout.

Custom **2×10 header footprints** were designed specifically to match the pin pitch and physical layout of the SIMCom cellular module, ensuring mechanical compatibility and reliable electrical connections. Special attention was given to power-related components, with bulk and decoupling capacitors placed close to their respective loads and wide traces used for the **VBAT** rail to safely handle higher current demands.

The PCB layout incorporated **top and bottom copper pours connected to ground**, along with over **100 ground-stitching vias**, to improve signal integrity, reduce noise, and enhance overall power stability. Manufacturer-recommended clearances and spacing rules were followed throughout the design to ensure manufacturability. Once completed, **Gerber files** were generated and verified, making the board ready for fabrication.

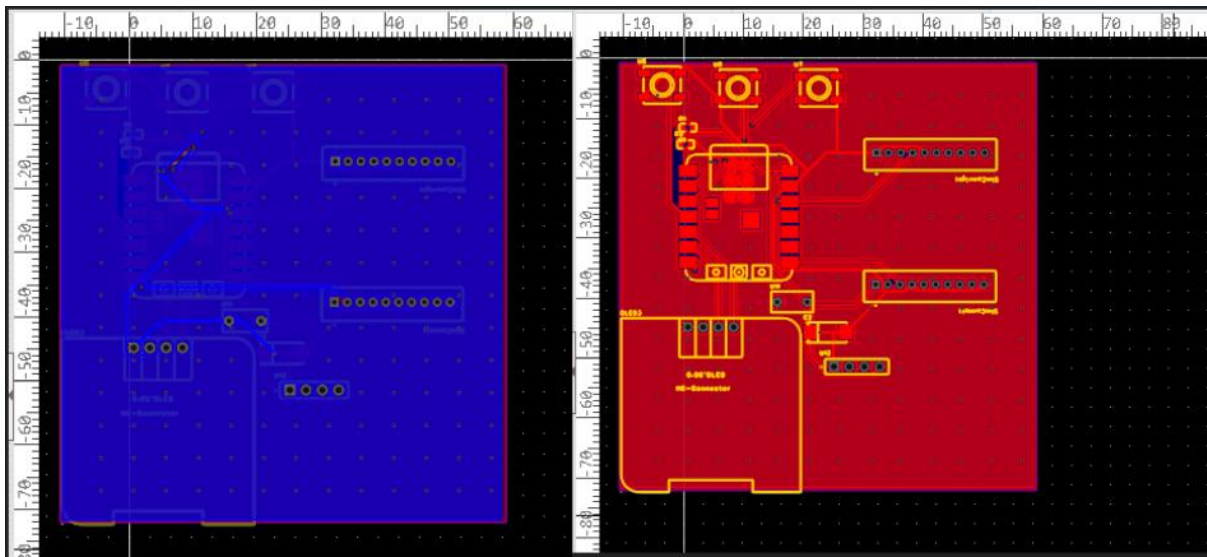


Figure 10. Top and bottom copper pours of the smartwatch PCB layout in EasyEDA, illustrating ground planes, signal routing, and component placement.

3.3 Soldering and PCB Assembly

The PCB assembly was carried out manually using soldering tin and a fine-tip soldering iron. Assembly followed a low-profile-to-high-profile approach to reduce mechanical stress and improve soldering accuracy.

Soldering began with the surface-mount components, starting with the 100 k Ω resistors and the decoupling and bulk capacitors (0.1 μ F and 100 μ F). These components were placed first to ensure stable power filtering and correct voltage sensing before adding larger parts.

Next, the OLED display and tactile push buttons were soldered, followed by the Seeed Studio XIAO ESP32-S3 module. Care was taken to ensure correct alignment and solid solder joints on all header pins.

Finally, the charging module, battery connections, and the SIMCom modem headers were connected to the PCB. After assembly, all solder joints were visually inspected to check for solder bridges or cold joints, and basic continuity checks were performed before powering the board. In addition, a **digital voltmeter was used to measure voltages across key points on the PCB**, including the battery terminals and power rails, to verify correct battery operation and confirm that the expected voltages were present before full system testing.

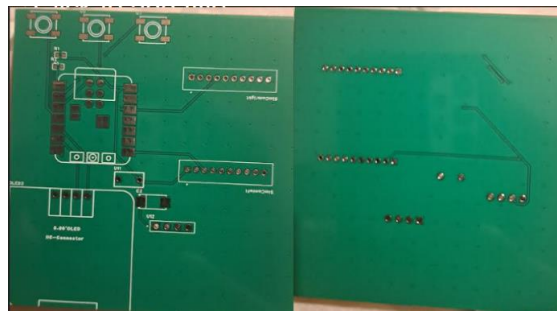


Figure 11. Manufactured smartwatch PCB (top and bottom views) showing completed routing, through-hole connections, and component footprints.



Figure 12. Manufactured smartwatch PCB (top view) with soldered components

3.4 Enclosure Development

A smartwatch enclosure was designed using **Autodesk Fusion 360** to house the PCB, battery, display, and associated components within a compact wearable form factor. The design was based on the finalised PCB dimensions to ensure accurate mechanical alignment and reliable assembly.

The enclosure includes internal mounting posts to securely support the PCB, accurately positioned openings for the tactile buttons, and a front display window aligned with the OLED screen.

Dedicated **USB-C cut-outs** were incorporated to allow external access to both the ESP32 XIAO programming port and the charging module. In addition, small side holes were included to enable the attachment of wrist straps.

The final enclosure was **3D printed using facilities at Maynooth University**, and the assembled electronics were successfully fitted inside the case. This validated the mechanical design and confirmed compatibility between the PCB layout, external interfaces, and the physical constraints of a smartwatch enclosure.



Figure 13. 3D printed smartwatch case (top and side views) with soldered PCB

3.5 Firmware Progress

In this stage, the firmware was built step by step to make sure each part of the smartwatch worked before combining everything. Development started on the XIAO ESP32-S3 in the Arduino IDE. Basic upload tests and Serial Monitor prints were used to confirm the board was powered correctly and the code was running as expected.

The OLED display was added next so the watch could show messages without relying only on the laptop. The I²C bus was checked with an I²C scanner to confirm the OLED address (0x3C), and then

the SSD1306 library was used to initialise the screen. Simple status text such as “Booting...”, “Ready”, and error messages was displayed to support debugging.

After that, the buttons were programmed. The pins were set as input pull-ups and a simple debounce was added to avoid multiple triggers from one press. Each button was linked to a preset, colour-coded message (for example “Ready in 5 minutes”, “Not well”, and “SOS”), and the OLED was updated to confirm which button was pressed and what action was happening.



Figure 14. ESP32-S3 Smartwatch Prototype During SMS and GPS Functionality Testing

Next, the SIMCom modem was integrated using UART and AT commands. Helper functions were written to send commands, read responses, and handle timeouts. At first the modem did not respond reliably, but once the correct TX/RX pin definitions for the XIAO ESP32-S3 were used, the modem consistently replied to AT with OK. The firmware then checked SIM status, signal strength, and network registration, showing messages like “Waiting network...” on the OLED while retrying.

Once the modem was registered on the network, SMS sending was implemented in text mode. The code sends the selected preset message to the parent’s phone number and confirms success using both modem responses and real SMS reception. Finally, GPS was added by enabling GNSS and reading location data from the modem. The coordinates were converted into decimal format and used to generate a Google Maps link, which was included in the SMS.

At the end of testing, the full MVP flow worked consistently: press a button → show progress on the OLED → connect to the network → get GPS → send an SMS with a Google Maps link to the parent’s phone.

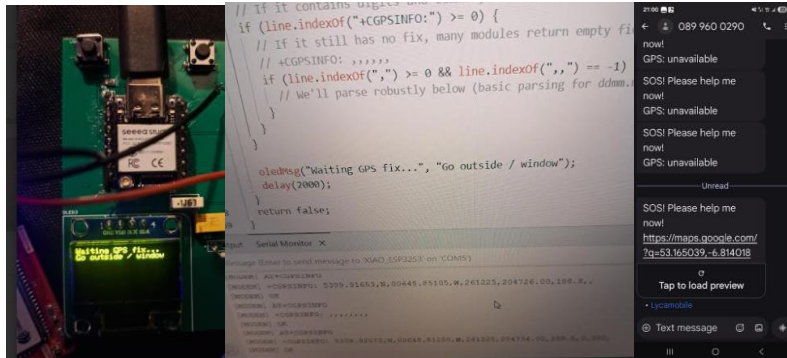


Figure 15. Successful SOS SMS Transmission with GPS Location Link Displayed on Mobile Phone

4 Conclusions and future work

This project successfully demonstrated a working **pre-teen safety smartwatch MVP** using the XIAO ESP32-S3, OLED display, push buttons, battery/charging subsystem, and a cellular modem for SMS + GPS location sharing. The system was developed in stages: first validating the ESP32, OLED (I²C), button inputs, and battery sensing, then transferring the design into a custom EasyEDA PCB and assembling it by hand.

A key outcome was reaching the full end-to-end flow: **press a button → show progress on the OLED → connect to the mobile network → obtain GPS coordinates → send an SMS with a Google Maps link** to the parent's phone.

This confirms that the overall hardware + firmware approach is suitable for a low-cost safety wearable prototype, while also highlighting the importance of correct power design, reliable soldering, and robust modem integration for real-world use.

Future improvements would focus on reliability, size reduction, and a more polished wearable design:

- **Make the smartwatch much smaller** by redesigning the PCB for a compact layout and using lower-profile components.
- **Improve PCB placement and positioning**, so parts (especially modem/power routing) do not disturb or interfere with other components mechanically or electrically.
- **Fix button reliability** (some buttons occasionally do not respond). This could be improved through better switch footprint alignment, stronger solder joints, improved debouncing, and possibly hardware filtering.

- **Modem integration refinement:** during development, the small modem approach had to be replaced with the larger SIMCom A7670E, and the firmware was updated so GPS data could be included correctly in the SMS. A future revision should select one final modem early and design the PCB + firmware around it from the start.
- **More testing in real conditions,** including battery life checks, network coverage tests, and repeated SOS/message trials, to confirm stable operation outside the lab environment.

Glossary of terms

AT Commands: Text-based commands (“Attention Commands”) used to control GSM/LTE modems such as SIMCom modules for actions like sending SMS, checking network status, or obtaining GPS data.

ESP32-S3 (XIAO ESP32-S3): A low-power microcontroller with integrated Wi-Fi and Bluetooth, used as the main controller in the smartwatch to manage inputs, display, and communication.

I²C (Inter-Integrated Circuit): A serial communication protocol using two lines (SDA and SCL) to connect low-speed peripherals such as OLED displays.

LTE (Long Term Evolution): A wireless broadband communication standard used by the SIMCom module to send SMS and data over cellular networks.

ON_OFF Pin: A control pin on the SIMCom module used to power the modem on or off by pulling the pin low for a specified duration.

SIMCom Module: A cellular communication module used to handle LTE connectivity, SMS transmission, and GPS functionality.

TX / RX: TX sends data out; RX receives data in. They must be cross-connected (ESP TX → Modem RX, ESP RX → Modem TX).

UART (Universal Asynchronous Receiver-Transmitter): A serial communication interface used to exchange data between the ESP32 and the SIMCom module.

VBAT: The main power input pin for the SIMCom module, typically requiring a stable voltage around 3.8–4.2 V.

References

Espressif Systems (2023) *ESP32-S3 Technical Reference Manual*. Espressif Systems. Available at:

https://www.espressif.com/sites/default/files/documentation/esp32-s3_technical_reference_manual_en.pdf

(Accessed: 20 December 2025).

NimbeLink Corp. (2022) *NL-SW-LTE-QBG95 Skywire Embedded Modem Datasheet*, Rev. 11.

NimbeLink Corporation. Available at:

<https://www.digikey.com/en/htmldatasheets/production/8068932/0/0/1/nl-sw-lte-qbg95>

(Accessed: 20 December 2025).

Appendix

Arduino code

```
#define MODEM_RX_PIN D7
#define MODEM_TX_PIN D6
#define BTN1_PIN D1
#define BTN2_PIN D2
#define BTN3_PIN D3

void setup() {
  Serial.begin(115200);
  Serial1.begin(115200, SERIAL_8N1, MODEM_RX_PIN, MODEM_TX_PIN);

  pinMode(BTN1_PIN, INPUT_PULLUP);
  pinMode(BTN2_PIN, INPUT_PULLUP);
  pinMode(BTN3_PIN, INPUT_PULLUP);

  Wire.begin();
  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  oledMsg("Smartwatch MVP", "Ready");

  sendAT("AT", "OK", 2000);
  sendAT("AT+CGNSSPWR=1", "OK", 3000); // Enable GPS
}

void loop() {
  if (digitalRead(BTN3_PIN) == LOW) {
    double lat, lon;
    String msg = "SOS! Please help me now!";
    if (tryGetGPS(lat, lon)) {
      msg += "\nhttps://maps.google.com/?q=" +
        String(lat,6) + "," + String(lon,6);
    }
    sendSMS(PARENT_PHONE, msg);
  }
}
```

Only a shortened firmware excerpt is shown above to illustrate the core functionality. The complete and fully commented Arduino source code is available at the link below:

[EE423 Arduino code - Smartwatch MVP \(XIAO ESP32-S3 + SIMCom A7670E + OLED\)](#)

Risk Assessment

Department of Electronic Engineering, Maynooth University

Health and Safety Risk Assessment and Guidelines.

Every project student or group of students must perform a health and safety risk assessment of their project before commencing any work.

This should be done in consultation with their project supervisor. The Risk Assessment form must be assessed, along with any required material such as Material Safety Data Sheets, and signed off by a member of the Health and Safety Committee.

The form is designed to identify any hazards that the proposed project might present to the student or people that may come into contact with the project.

Issues to consider include

- Hazardous materials, for example flammable materials, epoxies, cements, paints, aerosols, carcinogens, poisons, toxins...
- Working alone, at height, in confined space, in a dangerous area
- Working with electricity
- Working with machines and tools
- Working with bio materials
- Working with heat or flame (e.g. soldering or hot air guns)
- Working with lasers
- Working with heavy or large items
- Working with fast spinning objects (e.g. fans or propellers)
- Disposable / Recycling of hazardous materials eg batteries, plastics, metal and those mentioned above.

Where risks are identified, the student must discuss how the risk will be managed by listing what controls are in place and what further controls may be required.

The Risk Assessment Form should be included in your project report.

The Risk Assessment Form should be available for when requested by any staff member.

If this form is NOT available, then all work on the hardware aspects of your project MUST STOP until the form is signed off.

H&S committee members are: Mr. Andrew Meehan, Mr. John Maloco, Ms. Ann Dempsey, and Dr Bryan Hennelly.

I confirm, I have no medical condition that impedes working on my Project, and I will advise a Health & Safety Officer if this status changes.

Signed: _____

Department of Electronic Engineering, Maynooth University

PROJECT RISK ASSESSMENT FORM Rev. 3

NAME AND STUDENT NUMBER: Destiny Nosa - 22775725		PROJECT NAME: Pre-Teen Safety Smartwatch
SUPERVISOR: Andrew Meehan		PROJECT LOCATION: Maynooth University
BRIEF DESCRIPTION OF PROJECT Design, assembly, and testing of a low-voltage smartwatch prototype using an ESP32-S3, OLED display, buttons, and a cellular module messages with GPS location.		
Hazards, Risk [High(H) Medium (M) Low (L)], and Control Measures		
HAZARD	Risk	Controls
Soldering iron burns	M	Use temperature-controlled iron, avoid contact, allow cooling before handling.
Solder fumes	L	Work in a well-ventilated area.
PCB short circuits	M	Inspect solder joints, check continuity with multimeter before powering.
Li-PO battery misuse	M	Use correct charging module, avoid short circuits and over-voltage.
Component damage	M	Verify pinouts using datasheets and test subsystems individually.
Identified risks should be discussed with your supervisor and a safe system of work agreed. A more in-depth risk assessment may be required after initial review. Do not proceed until form is signed off.		
WEEE Compliance & Recycling: Indicate below how you intend to manage and dispose of your project's material during and post project work.		
Electronic components will be reused where possible. Any damaged electronics and batteries will be disposed of through approved WEEE recycling facilities.		
Further Controls Required		
SIGNATURE OF STUDENT: _____		DATE: 26/12/25
SIGNATURE OF SUPERVISOR: _____		DATE: _____
DEPT HEALTH AND SAFETY COMMITTEE MEMBER: _____		DATE: _____